PRISM-1R ARCHITECTURE BLUEPRINT

Hierarchical implementation view: package -> compute chiplet -> packet streaming multiprocessor

REFERENCE PACKAGE

Compute Chiplet C0
16 PSMs | 4 quadrants
8 transform | 4 solver

Compute Chiplet C1
16 PSMs | 4 quadrants
8 transform | 4 solver

Compute Chiplet C2
16 PSMs | 4 quadrants
8 transform | 4 solver

Compute Chiplet C3
16 PSMs | 4 quadrants
8 transform | 4 solver

UCIe package fabric

I/O + Control Die
RISC-V control | global scheduler
PCIe / CXL | security | telemetry

HBM4 0

HBM4 1

HBM4 2

HBM4 3

4 x HBM4 | reference aggregate > 11.2 TB/s

detail

ONE COMPUTE CHIPLET

Dual NoC: C0/C1 control + C2/C3 packet/reduction

Q0
4 PSMs
4 MiB shared SRAM
2 transform engines
1 solver engine

Q1
4 PSMs
4 MiB shared SRAM
2 transform engines
1 solver engine

Q2
4 PSMs
4 MiB shared SRAM
2 transform engines
1 solver engine

Q3
4 PSMs
4 MiB shared SRAM
2 transform engines
1 solver engine

hierarchical certificate aggregation + cross-quadrant traffic

Exact arithmetic cluster
modular | CRT | wide certificate checks

UCIe Router

detail

ONE PACKET STREAMING MULTIPROCESSOR

Packet front end
command + descriptor fetch

Dependency + generation
scoreboard

TRANSFORMATION PLANE
representation routing

generation-aware handoff

RESOLUTION PLANE
split | coarsen | refine

2 MiB PSM SCRATCHPAD | 16 x 128 KiB banks
P packets | F factors/transforms | M metadata | R residual/certificate

EG0
16x16 MMA
+ vector

EG1
16x16 MMA
+ vector

EG2
16x16 MMA
+ vector

EG3
16x16 MMA
+ vector

Projector /
polyphase DMA

Transform
port

Solver
port

Residual / certificate
accumulator

Commit / replay
controller

certificate failure -> local refine/replay

Execution rule: jointly choose representation, resolution, placement and precision

Commit rule: publish only after certificate acceptance and generation validation

Preserve structure | execute locally | refine only where evidence requires it